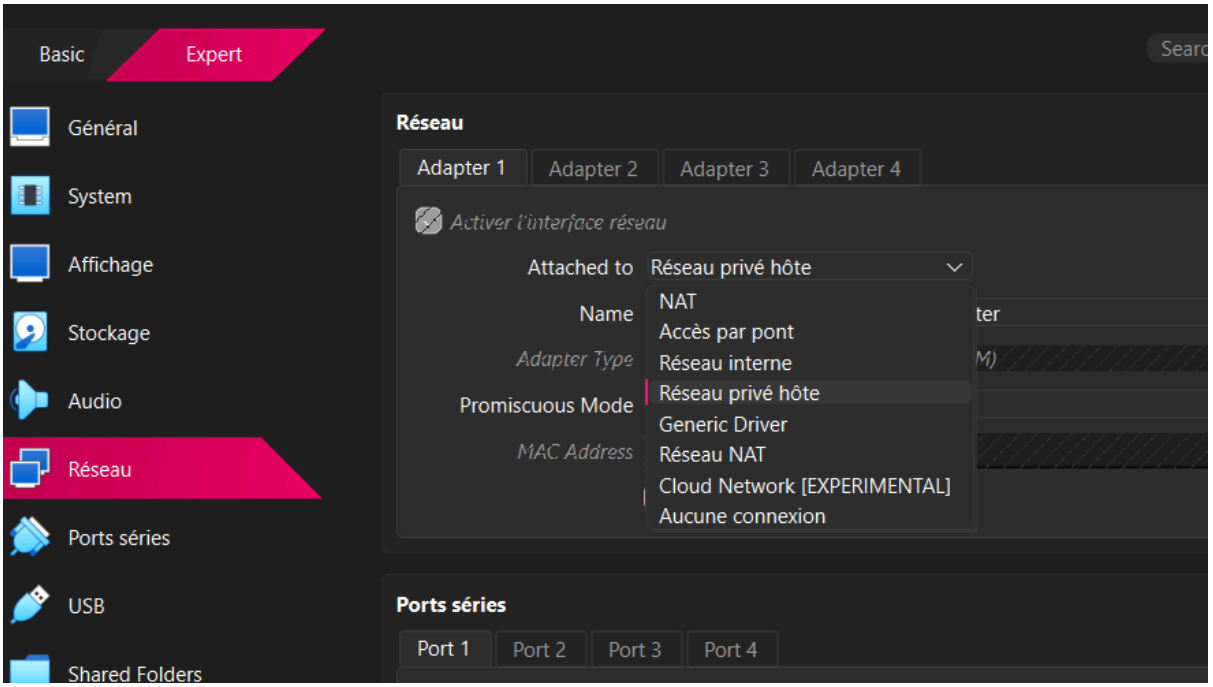
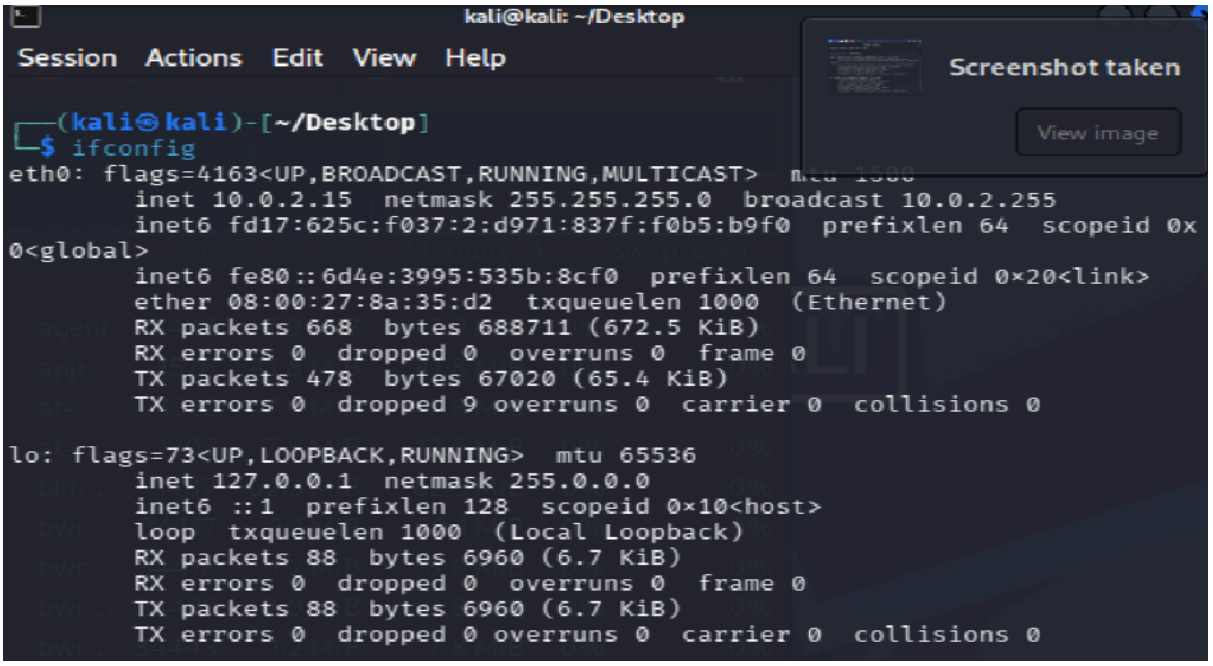
Dans cette partie, nous avons préparé l'environnement de travail en installant une machine virtuelle Kali Linux ainsi qu'une machine cible vulnérable Metasploitable2.

Kali Linux est utilisé comme machine attaquante contenant des outils de sécurité, tandis que Metasploitable2 est une machine volontairement vulnérable destinée à tester les techniques d'analyse.

Les deux machines sont exécutées sur VirtualBox.





- IP Kali (10.0.2.15)

Scan réseau

nmap 192.168.1.0/24

```
┌──(kali㉿kali)-[~/Desktop]
└─$ nmap 192.168.56.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-06 18:45 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disa
bled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.1
Host is up (0.00035s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT     STATE SERVICE
8081/tcp open  blackice-icecap
MAC Address: 0A:00:27:00:00:07 (Unknown)

Nmap scan report for 192.168.56.100
Host is up (0.00016s latency).
All 1000 scanned ports on 192.168.56.100 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:60:F6:03 (Oracle VirtualBox virtual NIC)
```

ça détecte les machines (dont Metasploitable)

Scan OS

nmap -O

```
┌──(kali㉿kali)-[~/Desktop]
└─$ nmap -O 192.168.56.102
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-06 18:46 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disa
bled. Try using --system-dns or specify valid servers with --dns-servers
Note: Host seems down. If it is really up, but blocking our ping probes, tr
y -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.52 seconds

┌──(kali㉿kali)-[~/Desktop]
└─$
```

Scan tous les ports

nmap -p-

```
┌──(kali㉿kali)-[~/Desktop]
└─$ nmap -p- 192.168.56.102
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-06 18:47 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disa
bled. Try using --system-dns or specify valid servers with --dns-servers
Note: Host seems down. If it is really up, but blocking our ping probes, tr
y -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.47 seconds

┌──(kali㉿kali)-[~/Desktop]
```

Vérifier ports 22 et 443

nmap -p 22,443

```
┌──(kali㉿kali)-[~/Desktop]
└─$ nmap -p 22,443 192.168.56.102
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-06 18:49 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disa
bled. Try using --system-dns or specify valid servers with --dns-servers
Note: Host seems down. If it is really up, but blocking our ping probes, tr
y -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.48 seconds

┌──(kali㉿kali)-[~/Desktop]
└─$
```

.

PARTIE 3 — NESSUS (VULNERABILITY SCANNER)

Installation

Dans Kali :

cd /home/kali/Downloads
sudo dpkg -i Nessus-*.deb

```
┌──(kali㉿kali)-[~/Downloads]
└─$ ls
Nessus-10.11.3-ubuntu1604_amd64.deb

┌──(kali㉿kali)-[~/Downloads]
└─$ sudo dpkg -i Nessus-*.deb
Selecting previously unselected package nessus.
(Reading database… 426006 files and directories currently installed.)
Preparing to unpack Nessus-10.11.3-ubuntu1604_amd64.deb…
Unpacking nessus (10.11.3)…
Setting up nessus (10.11.3)…
HMAC : (Module_Integrity) : Pass
SHA1 : (KAT_Digest) : Pass
SHA2 : (KAT_Digest) : Pass
SHA3 : (KAT_Digest) : Pass
TDES : (KAT_Cipher) : Pass
AES_GCM : (KAT_Cipher) : Pass
AES_ECB_Decrypt : (KAT_Cipher) : Pass
```

Démarrer service

sudo systemctl start nessusd.service

```
┌──(kali㉿kali)-[~/Desktop]
└─$ sudo systemctl start nessusd.service
[sudo] password for kali:

┌──(kali㉿kali)-[~/Desktop]
└─$ sudo systemctl status nessusd.service
● nessusd.service - The Nessus Vulnerability Scanner
     Loaded: loaded (/usr/lib/systemd/system/nessusd.service; disabled; pr>
     Active: active (running) since Mon 2026-04-06 18:50:33 EDT; 13s ago
 Invocation: 412b896fe4e54b268bcfae84a7a8245f
   Main PID: 93863 (nessus-service)
      Tasks: 14 (limit: 2118)
     Memory: 195.3M (peak: 195.3M)
        CPU: 7.436s
     CGroup: /system.slice/nessusd.service
             ├─93863 /opt/nessus/sbin/nessus-service -q
             └─93865 nessusd -q

Apr 06 18:50:33 kali systemd[1]: Started nessusd.service - The Nessus Vuln>
Apr 06 18:50:33 kali nessus-service[93863]: nessus-service [93863][INFO] :>
lines 1-14/14 (END)
```

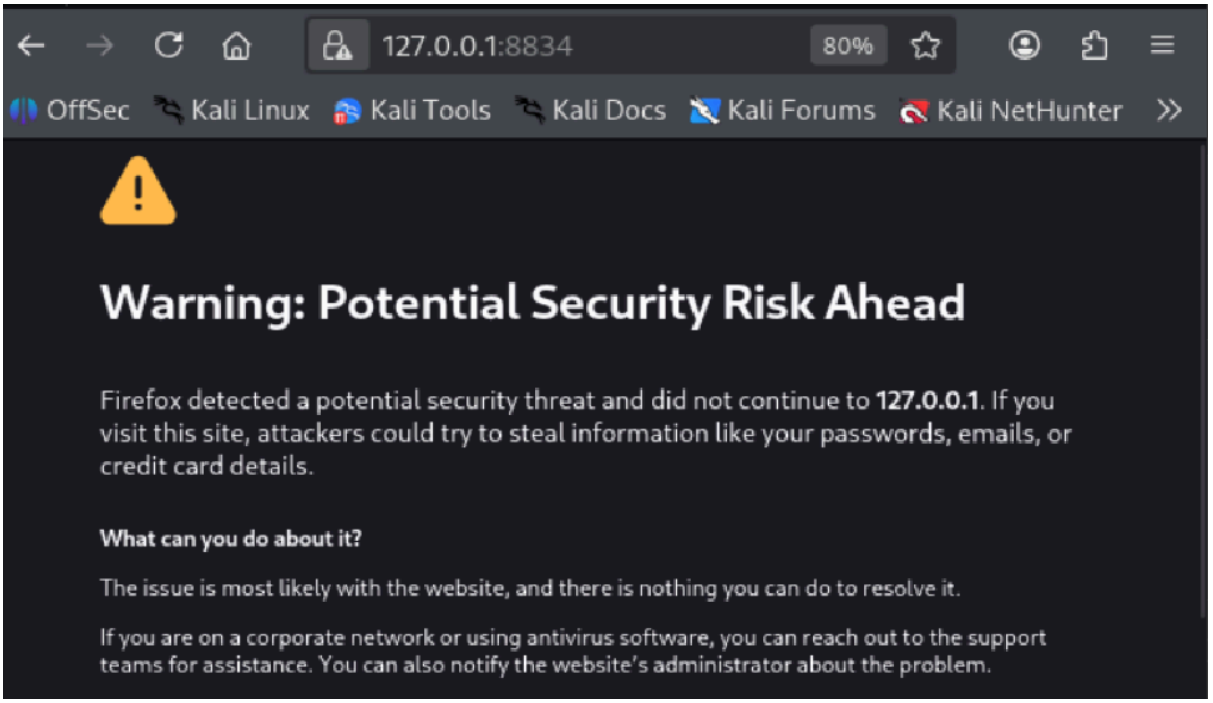
Vérifier

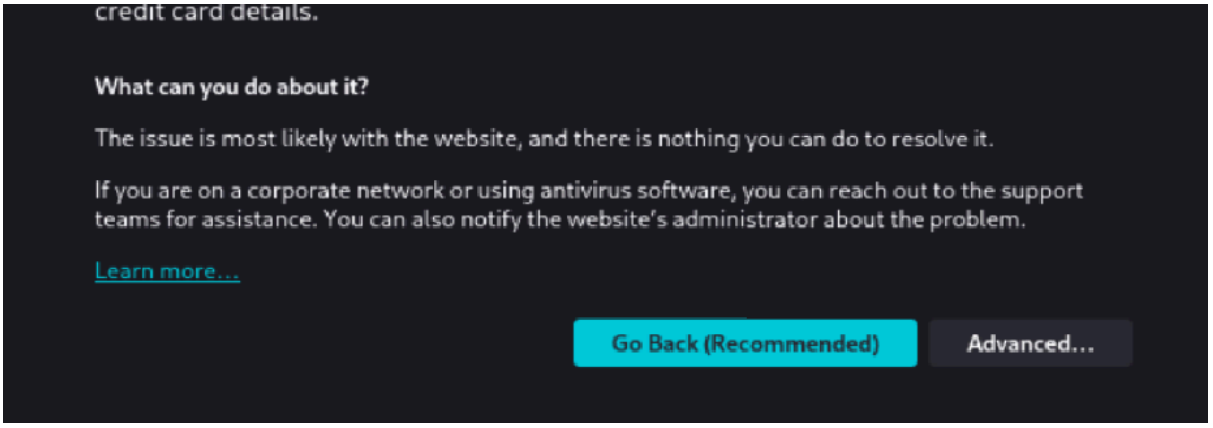
sudo systemctl status nessusd.service


```
┌──(kali㉿kali)-[~/Desktop]
└─$ sudo systemctl start nessusd.service
[sudo] password for kali:

┌──(kali㉿kali)-[~/Desktop]
└─$ sudo systemctl status nessusd.service
● nessusd.service - The Nessus Vulnerability Scanner
     Loaded: loaded (/usr/lib/systemd/system/nessusd.service; disabled; pr>
     Active: active (running) since Mon 2026-04-06 18:50:33 EDT; 13s ago
 Invocation: 412b896fe4e54b268bcfae84a7a8245f
   Main PID: 93863 (nessus-service)
      Tasks: 14 (limit: 2118)
     Memory: 195.3M (peak: 195.3M)
        CPU: 7.436s
     CGroup: /system.slice/nessusd.service
             ├─93863 /opt/nessus/sbin/nessus-service -q
             └─93865 nessusd -q

Apr 06 18:50:33 kali systemd[1]: Started nessusd.service - The Nessus Vuln>
Apr 06 18:50:33 kali nessus-service[93863]: nessus-service [93863][INFO] :>
lines 1-14/14 (END)
```


Ouvrir navigateur

https://localhost:8834





Setup

- Choisir : **Nessus Essentials**
- Créer compte
- Entrer licence (email)

PARTIE 4 — SCAN AVEC NESSUS

1. Scans → New Scan
2. Basic Network Scan
3. Target → IP Metasploitable

4. Launch

Nessus va afficher :

- Vulnérabilités
- Niveau : Low / Medium / High / Critical